

Diabetes Prediction using Neural Networks

Artificial Neural Network (ANN) Report

1. Data Preparation

This project uses the Pima Indians Diabetes Dataset (diabetes.csv): 768 patient records with 8 medical attributes (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) and a binary target, Outcome (1 = diabetes, 0 = no diabetes).

Dataset summary: 768 rows, 9 columns. No literal missing (NaN) values - but the dataset has a well-known quirk: several medical measurements use 0 as a placeholder for a value that was never recorded, even though 0 is biologically impossible for a living patient (e.g. 0 blood pressure or 0 BMI).

Column	Zero values	% of data
Glucose	5	0.7%
BloodPressure	35	4.6%
SkinThickness	227	29.6%
Insulin	374	48.7%
BMI	11	1.4%

Cleaning decision: Since Insulin and SkinThickness have very high zero-rates, dropping those rows would discard nearly half the dataset. Instead, the 0s in these five columns were replaced with the median of the real (non-zero) values in that column - a standard, robust approach for this well-known dataset issue.

Class balance: 500 patients with no diabetes vs. 268 with diabetes (~65% / 35%) - a moderate imbalance worth keeping in mind when interpreting accuracy later.

Feature/target separation: X = the 8 medical attributes, y = Outcome.

Train-test split: 80% train (614 patients) / 20% test (154 patients), stratified to preserve the class balance in both sets.

Feature scaling: StandardScaler was applied, fit only on the training data and then used to transform both train and test sets - this standardizes every feature to mean 0, standard deviation 1, which neural networks need since raw features like Glucose (0-200) and DiabetesPedigreeFunction (0-2.5) are on very different scales.

2. Building the Artificial Neural Network

Layer	Neurons	Activation	Role
Input	8	-	One neuron per medical feature
Hidden 1	16	ReLU	Learns non-linear feature combinations
Hidden 2	8	ReLU	Compresses learned patterns further
Output	1	Sigmoid	Outputs probability of diabetes (0-1)

Why ReLU for hidden layers: ReLU ($\max(0, x)$) is computationally simple and avoids the vanishing-gradient issues that older activations like sigmoid/tanh have in deeper networks, making it the standard default for hidden layers.

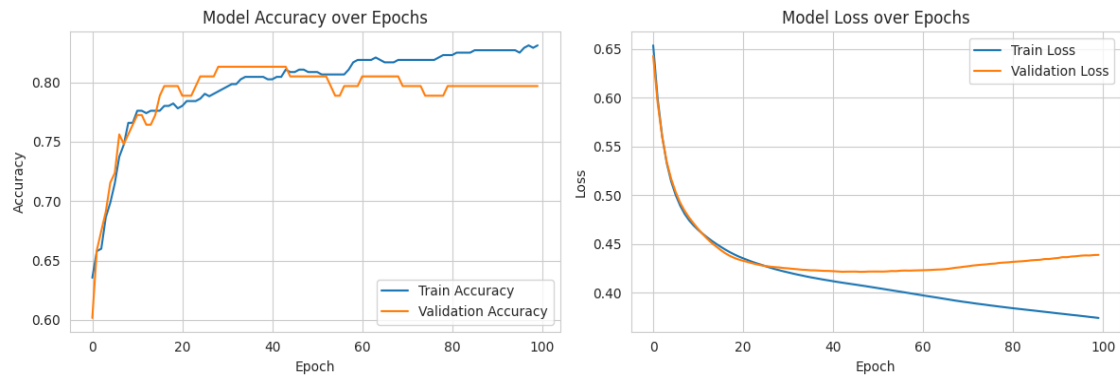
Why Sigmoid for the output layer: Sigmoid squashes output into a 0-1 range, which is interpreted directly as "probability of diabetes" - exactly what's needed for binary classification.

Compilation: Optimizer = Adam (adaptive learning rate, generally faster and more reliable convergence than plain gradient descent); Loss = Binary Crossentropy (the standard loss for binary classification, penalizing

confident wrong predictions more heavily); Metric = Accuracy (tracked during training).

3. Model Training

The model was trained for **100 epochs** with a batch size of 16, holding back 20% of the training data as a validation set to monitor for overfitting during training.

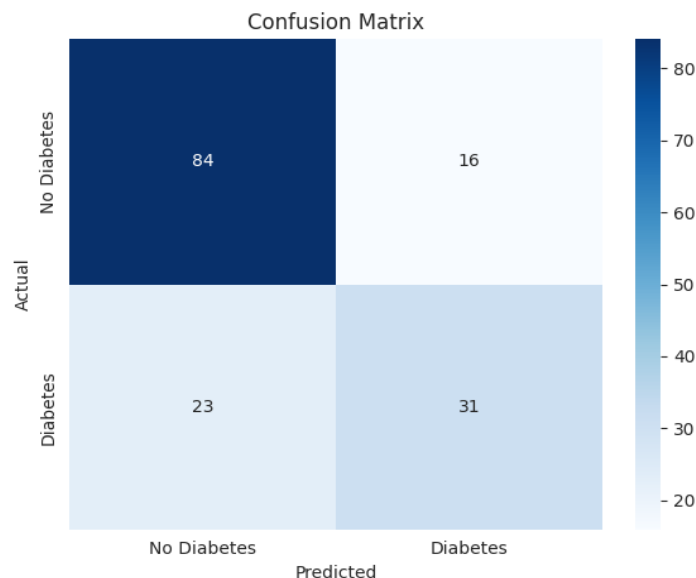


Final training accuracy: 83.10% Final validation accuracy: 79.67%

4. Model Evaluation

Predictions were generated on the held-out test set (154 patients) as probabilities, then converted to binary labels using a 0.5 threshold (probability > 0.5 = predicted diabetic).

Test Accuracy: 74.68%



Confusion matrix on the test set: 84 true negatives, 16 false positives, 23 false negatives, 31 true positives.

Metric	No Diabetes	Diabetes
Precision	0.79	0.66
Recall	0.84	0.57
F1-score	0.81	0.61
Support	100	54

Interpretation:

- The model correctly identifies 84 of 100 non-diabetic patients (84% recall for that class) but only 31 of 54 diabetic patients (57% recall) - meaning it misses (false-negatives) 23 diabetic patients entirely.
- Given the class imbalance noted in Part 1 (65% No Diabetes / 35% Diabetes), overall accuracy alone is a bit misleading - a model that always predicted 'No Diabetes' would already score ~65% without learning anything. The 74.68% accuracy here does represent real learning, but the lower recall on the Diabetes class is the more clinically important number to improve.
- In a medical screening context, false negatives (missed diagnoses) are generally more costly than false positives (a healthy patient flagged for further testing), so future work could focus on techniques like class weighting, oversampling the minority class, or adjusting the decision threshold below 0.5 to catch more true diabetes cases at the cost of some extra false alarms.

Assignment: Diabetes Prediction using Neural Networks - UST Gen AI Engineers Program.